Hybrid Model of Power Transformer Fault Classification Using C-set and MFCM – MCSVM

Ali Abdo, Hongshun Liu, *Member*, *IEEE*, Yousif Mahmoud, Hongru Zhang, Ying Sun, Qingquan Li, and Jian Guo

***Abstract*—This paper aims to increase the diagnosis accuracy of the fault classification of power transformers by introducing a new off-line hybrid model based on a combination subset of the et method (C-set) & modified fuzzy C-mean algorithm (MFCM) and the optimizable multiclass-SVM (MCSVM). The innovation in this paper is shown in terms of solving the predicaments of outliers, boundary proportion, and unequal data existing in both traditional and intelligence models. Taking into consideration the closeness of dissolved gas analysis (DGA) data, the C-set method is implemented to subset the DGA data samples based on their type of faults within unrepeated subsets. Then, the MFCM is used for removing outliers from DGA samples by combining highly similar data for every subset within the same cluster to obtain the optimized training data (OTD) set. It is also used to minimize dimensionality of DGA samples and the uncertainty of transformer condition monitoring. After that, the optimized MCSVM is trained by using the (OTD). The proposed model diagnosis accuracy is 93.3%. The obtained results indicate that our model significantly improves the fault identification accuracy in power transformers when compared with other conventional and intelligence models.**

***Index Terms*—Combination subset of set (C-set) method, modified fuzzy C-means (MFCM), optimizable multiclass-SVM (MCSVM), optimized training data (OTD).**

I. Introduction

POWER transformers are considered one of the most significant devices in power systems. Faults in power transformers primarily occur from cumulative electrical, mechanical, chemical, and thermal pressures resulting from their long-term operations [1]. These deficiencies lead to the decay of the internal segments of the transformer, such as windings, core laminations, and joints losses. As a result, gaseous products, such as Methane (CH_4), Ethan (C_2H_6), Ethylene (C_2H_4), Acetylene (C_2H_2), Carbon Dioxide (CO_2), Carbon Monoxide (CO) and Hydrogen (H_2), manifest themselves in parts per million (ppm) within the transformer's mineral oil. The concentrated level of the gases depends on different factors, such as temperature changing, transformer load, or the type of faults [2]–[4]. Multiple techniques, such as dissolved gas analysis (DGA), degree of polymerization measurements (DPM), gas chromatography (GC), mass chromatography (MC), were proposed to screen the insulation conditions and to detect faults in the transformer [3]. Among previous techniques, DGA is the most widely recognized method that can be used frequently to monitor the transformer's overall condition and provide speed up warnings of new developing faults [5]. DGA data interpretation can be conducted by using some diagnostic methods, such as the key gas methods [6], graphical methods [7], and ratio methods [8].

These techniques work on setting predefined criteria to excerpt features from gas data [9]. However, the collected gas concentrations or ratios obtained by the traditional methods may be incomplete; thus, it does not fit the predefined boundaries, which may cause different judgments when analyzing the same DGA data record. Moreover, conventional methods fail to provide the correct diagnoses when multiple faults, such as discharge and thermic defects, simultaneously occur. The unsettled explanation of these combined defects comes from the objective ambiguity, which can be related to the causal sequence between the transformer defects and the subjective boundary of the verification data set. This uncertainty prompts the decrease in the effectiveness and accuracy of these models. Eventually, traditional methods suffer from incorrect classification and inconsistent results for similar oil samples. Therefore, problems associated with these methods need to be solved for reliable fault diagnosis systems [10]. Due to the shortcomings associated with classic models, scientists have changed their focus toward intelligent techniques (AI), such as artificial neural networks NN [5], cognitive system (CS) [11], combined deep neural networks (CDNN) [12] and extreme learning machines (ELM) [13]. These techniques reveal an incredible improvement in fault analysis in terms of attaining high accuracy and efficiency. However, these methods also have limitations, such as the complex acquisition of knowledge from data sets, which require a substantial and complete amount of data samples to obtain high accuracy and reliable performance. Some of these methods are sensitive to parameter selection and need time and memory space to properly perform simulations. Even though these previously mentioned methods may achieve good accuracy, they could fall into the problem of outliers if the data is not correctly handled. Outliers in the data set have unwanted effects; for instance, they can destroy the connections between the exposure and outcome due to the creation of model biasing. Outliers can also cause the error

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variance of the model to increase and create deficient effects by disconnecting the cluster centers from their optimum level. As a result, the quality of the training data set and the accuracy of the classification model will be reduced [14]. Moreover, these methods might also suffer from the effects of unbalanced data sets, in which the model will see more records from the majority classes than minority classes. Therefore, the classifier will not be sufficiently trained, and most of the obtained patterns will also be from majority classes, which will cause the samples of the minority classes to be ignored. This effect will greatly affect the training process and eventually cause a major reduction in the model performance accuracy when the final classification is performed [15]. Thus, the complication of these methods makes it extremely difficult to implement them practically in electrical utilities [16].

Accordingly, this paper introduces an advanced off-line model to reduce some of the shortcomings previously mentioned in classical and intelligent methods. The main focus of this paper is to analyze the abnormal data samples. This paper's innovation appears in the data preparation step in terms of applying the combination subset of the set (C-set) method and modified fuzzy C-means algorithm MFCM to solve the problems of unbalanced data sets outliers, predefined boundaries, and the large size of data samples. The process starts by separating the normal samples from abnormal samples of the original data set using the decision rules. After that, the abnormal data samples will be normalized and split into training and testing sets. Then, the principle of the C-set method is enforced within the transformer historical training data set to partition it into different subsets that contain highly relevant information. After that, a modified fuzzy C-means (MFCM) algorithm is separately applied to the subsets. This algorithm will remove outliers from the training data and minimize each subset's size by combining highly similar data into a cluster and assigning fuzzy membership values for each data point. The cluster centers obtained from each subset will be combined to obtain the optimized training data set (OTD). The obtained training data set will be a balanced set, in which all types of faults will have an equal number of samples in the training data set. Finally, this balanced data will be used to train the optimizable multiclass-SVM algorithm (MCSVM). After training the optimized MCSVM using the data obtained from literature, the model was exported to MATLAB 2019b workspace to be evaluated by using real testing data obtained from the Zibo electrical power company. The overall accuracy obtained after testing our model is 93.3%. Accordingly, our model provides faster estimation speed and successfully extracts the most extreme predictive values from any defect within the given data set. The performance evaluation carried out in this paper shows promising steps in extracting and simplifying the meaningful values of larger data sets down to smaller groups in power transformer fault diagnosis. Our proposed model (C-set & MFCM - MCSVM) obtained a remarkable performance in classifying transformer faults when compared with other classical and intelligence methods, such as (KNN) (K-nearest neighborhood), SVM cubic, Naive Bayes, Decision Tree, Fine Gaussian SVM, Neural Networks (NN), IEC 60599, Rogers ratio, Duval pentagon, and Duval triangle.

The remainder of this paper is organized as follows: Section II explains the fault classification model of faults in power transformers regarding their theoretical background and methodology. It starts by providing a general overview of the fault patterns evolving inside the PT. Then it discusses the mathematical foundations of the C-set method, MFCM, and MCSVM algorithms. After that, it explains the C-set-MFCM & MCSVM model details, the data processing, the separation process, and discusses the final results obtained after testing the proposed model. Section III contains a discussion that compares our model's performance accuracy with other classical and intelligence model accuracies. Finally, Section IV provides the conclusion of this paper.

II. Power Transformer Fault Diagnosis Model Using (C-set & MFCM - MCSVM)

A. Theoretical Background

1) A General Overview of the Fault Pattern Evolving Inside the PT Oil

All power transformers produce gases when they are operating; however, when the production level of these gases exceeds certain limits, both electrical and thermal faults start to emerge. The identification of these faults and their severity level is presented by analyzing the production level of the gases as follows:

It appears that large amounts of H_2, CH_4, and small quantities of C_2H_4, C_2H_6, manifest themselves at a temperature range from 150°C to 500°C when mineral oil starts to decompose. The amount of these gases at this level is taken as a sign of low-temperature overheating defects. As the temperature of the transformer increases to medium state, it is noted that the amount of H_2 exceeds that of CH_4, and a significant amount of C_2H_6 and C_2H_4 appears in the oil. As the temperature reaches a higher state, which could be a possible indication of high-temperature faults, it is observed that the quantities of C_2H_4, H_2 increases, and traces of C_2H_2 may be formed. Moreover, the evolvement of electrical faults, such as PD and LED, can also be indicated by measuring the concentration level of H_2 and other gases. For this type of fault, the H_2 is the dominant gas, with a reduction in the amount of CH_4 and a trace of C_2H_2. As the intensity level of discharge increases, the concentration level of C_2H_2 and C_2H_4 also increases. In addition, when the intensity level of electric discharge arrives at the arcing level or the proportion of continuous discharges that generates temperature in the range between 700°C to 1800°C, the quantity of C_2H_2 becomes more dominant [6].

2) The Combination Subset of Set METHOD(C-SET)

A combination of set G can be simply defined as the selection of a random non-negative integer z of h objects in the G set without redundancy. This type of picking produces a subset z from G, such that $z \subseteq G$; $G = [h\text{-}elements]$. If set G contains three items $\{d, u, h\}$, then $\{d, u\}$, $\{d, h\}$, $\{u, h\}$ are the three subsets for each 2-combinations of G. The number of z-subsets of h-items is denoted by $\binom{h}{z} = 0$ if $z > h$. Accordingly, the basic equation for the total number

of E groups obtained from set G is given in the following equation [17]:

$$E = \binom{h}{z} = \frac{h!}{z!(h-z)!} \tag{1}$$

where E represents the total number of classes; $h!$ represents the items number in G; $z!$ represents the selected classes number, and $(h-z)!$ represents the remaining items in the set.

3) Modified Fuzzy C-means Clustering Algorithm (MFCM)

Fuzzy C-mean (FCM) algorithm is a type of unsupervised clustering method that works on the principle of changing the classification quandaries into a mathematical optimization problem with constraints [18]. However, this method has some issues, such as local extrema that results from the increased sensitivity of the initial values. Therefore, different clustering results are obtained for the same data set in every iteration, which eventually reduces the clustering accuracy. Moreover, as the input data set increases, the membership function exhibits asymptotic points, and its monotonicity becomes more severe. Thus, the modified fuzzy C-means algorithm has emerged to overcome the problems associated with traditional FCM. This method has an exponential membership function, which removes the local extrema, and has very good monotonicity, which smoothly divides input space into clusters. This method has eliminated the sensitivity problem, and it became beneficial in practical engineering applications [19].

The objective function (OF) of MFCM is presented as follows:

$$J(U,V) = \sum_{i=1}^{N}\sum_{j=1}^{r} u_{ij}^{m} \mathrm{e}^{-\gamma d_{ij}^2} \tag{2}$$

The OF is minimized using the Lagrange multiplier, and the $J(U,V)$ function is differentiated to u_{ij} for (fixed v_j) and v_j for (fixed u_{ij}), accordingly. The equations of the cluster centers and membership functions appear as follows:

$$u_{ij} = \frac{s_{ij}}{\sum_{k=1}^{r} s_{ik}} = \frac{\prod_{h}^{s} \mathrm{e}^{-\gamma(c_{ih}-v_{jh})^2}}{\sum_{k=1}^{r}\prod_{h=1}^{s} \mathrm{e}^{-\gamma(c_{ih}-v_{kh})^2}} \tag{3}$$

$$v_j = \frac{\sum_{i=1}^{N} u_{ij}^{m} c_i}{\sum_{i=1}^{N} u_{ij}^{m}} \tag{4}$$

where r is the total number of clusters in the data set; N is the total number of samples in the set; V is the matrix of the cluster centers; d_{ij} represents the Euclidean distance between the jth cluster center and the ith data point $d_{ij}^2 = \|c_{ih} - v_{jh}\|^2$; $U = \{u_{ij}\}$ is the matrix membership; u_{ij} is the membership function of the jth class that has the ith sample belonging to it; v_j is the obtained final cluster centers; c_{ih} is the given data point given in the set; v_{jh} is the centroid of cluster j; $m \in [1, +\infty]$ is the weighing exponent factor that represents the clusters fogginess of the MFCM; γ represents the shape parameters that is used to adjust the sensitiveness of the sub-similarity function $S_{ijk} = \mathrm{e}^{-\gamma(c_{ih}-v_{jh})^2}$ of each gas feature. The $S_{ij} = \prod_{h}^{s} \mathrm{e}^{-\gamma(c_{ih}-v_{jh})^2}$ formula represents the comprehensive similarity function; s is the number of input features. The iteration process for (3) and (4) is repeated until the desired accuracy level is obtained.

4) Multiclass Support Vector Machine

Support Vector Machine (SVM) is a type of supervised classification algorithm used to solve quandaries with small samples, nonlinearity, local minima, and high dimensions. However, traditional $\mathrm{SVM_s}$ are considered binary classifiers that are not suitable for solving multiclass classification problems because of their high sensitivity to outliers and noises, which tend to become support vectors (SV) in the training stage [20]. Moreover, as the data set's dimension is expanded, the SVM algorithm becomes very slow in data processing. As a result, it is necessary to construct multiclass-SVM models, to exclude the impartibility sector and enhance the training accuracy and the speed of measurements using multiple combination policies. Multiclass-SVMs are categorized into different types, such as 1-vs-all, 1-vs-1, decision tree, and directed acyclic graph support vector machines DAG SVM. In this paper the MCSVM 1-vs-1 method is used. This method starts by taking two classes' samples from the v_{th} classes every time, and then an SVM is constructed for every possible combination of these two classes. Accordingly, the total SVM sub-classifications are obtained by using $v(v-1)/2$ [21]. The main idea of SVM is that nonlinear data samples consist of two categories/classes $\{v_i, y_i\}$, where $i = 1, 2, \cdots, N$; $v \in R^N$, $y \in \{-1, 1\}$; N is the quantity of data samples which needs to be mapped from its original dimension to a higher dimensional character space P by using the pre-selected nonlinear mapping function $\phi(v)$ so that the optimal hyperplanes can be obtained by maximizing the distance between the two groups of data samples through using the linear classification equation as follows:

$$D(v) = [w.\phi(v) + b] \tag{5}$$

where w is the typical vector of the final separating hyperplane, and b is constant. In general, the optimal hyperplane solution is obtained by solving the constraint optimization problem in (6) as follows:

$$\min \frac{1}{2}w^2 + C\sum_{i=1}^{N}\zeta_i \tag{6}$$

$$\text{Subject to: } y_i.[w.\phi(v) + b] + \zeta_i \geq 1;\ 1 \leq i \leq N;\ \zeta_i \geq 0$$

where: The regularization parameter C is the penalty factor that is used to identify the trade-off between empirical risk and the interval confidence; w is the width of the margin; ζ_i is the slack factor. The reduction of the previous convex optimization problem into dual quadratic optimization problems is performed by combining the method of Lagrange multipliers as follows:

$$\max Q(\delta) = \sum_{i=1}^{N}\delta_i - \frac{1}{2}\sum_{i,j=1}^{N}\delta_i\delta_j y_i y_j \phi(v_i).\phi(v_j) \tag{7}$$

$$\text{Subject to: } \delta_i \geq 0,\ 1 \leq i \leq N;\ \sum_{i=1}^{N}\delta_i y_i = 0$$

where δ represents the Lagrangian multiplier. Then, the nonlinear decision function is shown as:

$$D(v) = \mathrm{sign}\left(\sum_{i,j=1}^{N}\delta_i y_i(\phi(v_i).\phi(v_j)) + b\right) \tag{8}$$

From (8) the values of $\phi(v_i)$. $\phi(v_j)$ can be calculated using the kernel function $k(v_i, v_j)$. Accordingly, the $D(v)$ equation is described in terms of the kernel function:

$$D(v) = \text{sign}\left(\sum_{i,j=1}^{N} \delta_i y_i k(v_i, v_j) + b\right) \tag{9}$$

Compared with other kernels, it is noted that the linear kernel has the ability to obtain higher classification accuracies and make the computation time of the algorithm very fast. The multiclass-SVM linear function is obtained from the following equation:

$$K(v_i, v_j) = v_i^{\mathrm{T}} v_j \tag{10}$$

B. Methodology

According to the class reference of IEC60599, the DGA data obtained from oil-immersed power transformers consists of two categories that include the normal and abnormal data samples [22]. Therefore, after collecting historical data from the inspected power transformers, the next step will be to interpret and analyze these data samples by using our model.

The proposed model in Fig. 1 starts by pre-processing the collected data samples using the proportion of gases method. After that, decision rules are implemented to separate the normal from abnormal data samples. Then, the train and test split method is implemented to divide the abnormal data into two groups, one for training and the other for the final evaluation of the model. After that, the initial training data set will be partitioned into multiple classes without repetition using the C-set method. Later, every subset will be clustered using the non-parametric MFCM algorithm to reduce the size of each subset and the parameter estimation errors in the clustering process. After obtaining the optimum cluster centers from every subset, they can be combined to form the optimized training data set (balanced data set). This set will be used as an input to train the optimized MCSVM algorithm. Then, when the model is finally trained, the testing samples taken from the Zibo electrical power company can be used to evaluate the classification performance of the trained model and obtain the final diagnosis of power transformer faults.

1) Data Optimization

a) Data Pre-processing: The performance of C-set & MFCM and MCSVM can be made more accurate by performing specific preprocessing steps on the original data set. This paper has adopted the historical data set from literature, such as IEC 60599 publication and IEC TC 10 databases [19], [22], and the State grid power supply company in Zibo, China. This data set contains 239 field data (in Table AV) that includes five gases as input and two class categories as an output. One class is normal, and the other is abnormal data samples, which contain six types of faults that have been identified by the transformer's visual investigation after the service's fault. These faults are Partial Discharge (D_1∼PD), Low Energy Discharge (D_2∼LED), High Energy Discharge (D_3∼HED), Low-Temperature Overheating (D_4∼T_1), Medium Temperature Overheating (D_5∼T_2), High-Temperature Overheating (D_6∼T_3). Since the hydrogen value is considerably larger

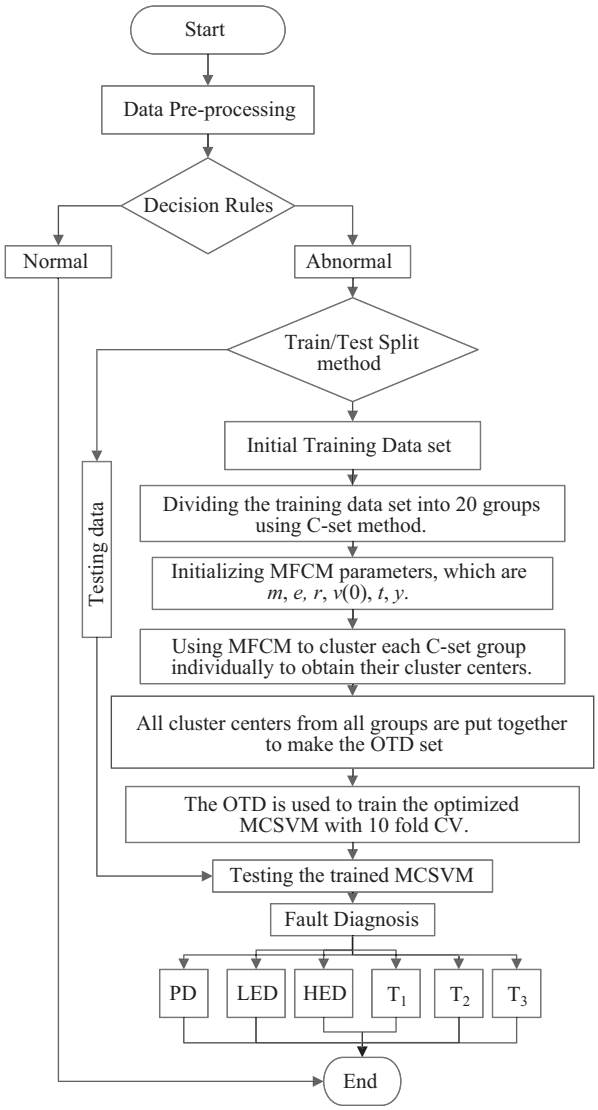


Fig. 1. The C-set & MFCM-MCSVM proposed Model.

when compared to the amounts of the other gases, we, therefore, take the proportion of H_2 to (H_2+ total hydrocarbons) and the ratio of each hydrocarbon to the Total Hydrocarbons ($To.Hy$) [19].

Where

$$To.Hy = \sum(\mathrm{CH_4} + \mathrm{C_2H_6} + \mathrm{C_2H_4} + \mathrm{C_2H_2}) \tag{11}$$

C. The Separation Process

The separation process starts by applying the decision rules to the original data set (before normalization) to separate the normal samples from abnormal samples. These rules are derived from the IEC 60599 standard and created based on the typical concentration levels of the dissolved gases in Table I; therefore, two rules are used in the separation [23].

Rule 1: if the value level of all gases in the sample is below the standard concentration level and the gas increase rates adopted by IEC 60599, the sample will be identified as a normal/healthy sample.

Rule 2: if at least one of the gases has a higher value than the standard value, the sample will be treated as a suspicious

TABLE I
RANGES OF 90% TYPICAL CONCENTRATION GAS LEVELS OF ALL TRANSFORMERS IN (PPM) ACCORDING TO IEC 60599 [21]

Dissolved Gases	All Transformers	No OLTC*	Communicating OLTC*
C_2H_2	–	3–50	60–280
H_2	50–150		
CH_4	30–130		
C_2H_4	60–280	*On Load Tap Changer (OLTC)	
C_2H_6	20–90		
CO	400–600		
CO_2	3800–14000		

sample and require further investigation. These investigations include examining the suspicious samples using the IEC 60599 standard methods, such as Duval triangle, Duval pentagon, key gas, and Roger's methods to obtain a preliminary diagnosis of the suspicious sample. The final diagnosis will then be based on the maximum number of votes obtained by IEC 60599 methods.

From these two rules, a programming code was developed and applied to all transformer data samples, and the obtained results were confirmed by experts. An example is introduced to demonstrate the working process of the decision rules. In this example, five raw samples are taken for investigation from the original data set and inputted into the decision rules programming code. The concentration level of these samples' gases is shown in Table II.

TABLE II
THE SAMPLES USED FOR TESTING THE DECISION RULES

No.	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Diagnosis
1	14.67	3.68	10.54	2.71	0.2	Normal
2	7.5	5.7	3.4	2.6	3.2	Normal
3	220	340	42	480	14	T_3
4	30	110	137	52	22.3	Normal
5	80	10	4	1.5	0	Normal

From the Table II, the concentration level of the input gases in the sample numbers $\{1, 2, 4, 5\}$ fall within the range or below the standard gas level in Table I; this means that Rule 1 is applicable and the final diagnosis output for these samples is the normal healthy state condition. As for sample number 3, the concentration level of $\{H_2, CH_4, C_2H_4\}$ are higher than the standard values shown in Table I; in this case, Rule 1 is not applicable, and the output of rule two means that this sample needs further investigation. The investigation is performed using four methods obtained from the IEC 60599 standard, which are used to obtain the diagnosis of sample number three, as shown in Table III.

TABLE III
THE FINAL DIAGNOSIS OF SAMPLE NUMBER THREE BASED ON MAXIMUM VOTING

Sample No.3	Rogers	Duval Triangle	Duval Pentagon	Key gas method	Final Diagnosis
	T_3	T_3	T_3	T_3	T_3

From Table III, it is noted that all IEC 60599 methods have produced the same diagnosis. Therefore, the final diagnosis based on the maximum voting obtained for sample three is the High temperature overheating fault (T_3).

A total of 239 samples in Table AV were investigated in the same way, and the final results after separation include 33 samples which were identified as normal and 206 samples as abnormal. The distribution of normal and abnormal data samples is shown in Table IV.

TABLE IV
DISTRIBUTION OF DGA DATA SAMPLES AFTER SEPARATION

Samples	Symbol	Samples Num	Total	All Samples
Normal Samples	N	33	–	239
Abnormal Samples	PD	20	206	
	LED	36		
	HED	58		
	T_1	33		
	T_2	31		
	T_3	28		

After that, the proposed fault classification model based on the C-set & MFCM- optimizable MCSVM method is implemented to classify the 206 abnormal data set as follows:

1- Train/Test Split Technique

By applying the train and test method, the 206 collected data samples have been divided into 176 samples for training (this data set is obtained from literature) and 30 samples for testing (this data set is obtained from the Zibo electrical power company), as shown in Table V:

TABLE V
DISTRIBUTION OF THE TRAINING AND TESTING DATA SAMPLES

Symbol	Type of Fault	Training set	Testing set
D1	PD	16	4
D2	LED	31	5
D3	HED	51	7
D4	T_1	29	4
D5	T_2	26	5
D6	T_3	23	5

2- Applying the C-set Method

The six types of faults mentioned in Table I were assigned random values from one to six as follows: $\{PD = 1, LED = 2, HED = 3, T_1 = 4, T_2 = 5, T_3 = 6\}$. Then (1) was applied to identify the number of subsets obtained from the original set. In this case, every subset will contain three types of faults. The reason for selecting the three fault combinations is that it gives us the maximum number of data samples obtained in the final set, which are 60 samples (the final balanced data set obtained after clustering). To clarify, the 176 data samples in the original set are divided into 20 groups; each contains a total of 3 faults according to (1). Where E is the total number of the groups; $h = 6$ is the number of defects (classes); $z = 3$ is the number of faults classes being chosen at a time without repetition. Table VI shows the 20 groups of fault combinations obtained by (1).

3- Modified Fuzzy C-means Clustering

Clustering algorithm-based approaches can observe outliers by investigating the relationship between data points and clusters. Naturally, an outlier can be characterized as a data point that belongs to a small and isolated cluster or does not belong to any cluster at all. Alternatively, it is assigned a low

TABLE VI
COMBINATION OF 20 GROUPS, EACH CONTAIN THREE TYPES OF FAULT

No.	Faults Combination	No.	Faults Combination	No.	Faults Combination
A	D1&D2&D3	G	D1&D3&D5	M	D2&D3&D4
B	D1&D2&D4	H	D1&D3&D6	N	D2&D3&D5
C	D1&D2&D5	I	D1&D4&D5	O	D2&D3&D6
D	D1&D2&D6	J	D1&D4&D6	P	D2&D4&D5
E	D1&D3&D4	K	D1&D5&D6	Q	D2&D4&D6
F	D2&D5&D6	L	D3&D4&D5	R	D3&D4&D6
S	D3&D5&D6	T	D4&D5&D6	–	–

membership value in which the distance between data points and the center of the cluster is higher than the minimum inter-clusters distance [24]. In this paper, the optimized training dataset is obtained by using the MFCM algorithm.

This algorithm is used to reduce the quantity of purified information in each class of the data set. In other words, the MFCM picks instructive delegate information for each class of the data set by applying the principle of data cleaning as follows:

- Each subset of the original set is clustered using MFCM. Therefore, the data samples assigned low membership values or not clustered correctly by the algorithm are considered outliers, and they were eliminated. This process's importance is to help prevent the similarity between the cluster centers of each subset and obtain the maximum separation between cluster centers. After elimination, the new subsets were clustered again while changing the values of m and y.
- This process is continuously repeated until the optimum cluster centers of each subset are obtained.

Subsequent to applying the MFCM algorithm, the following parameters in (3) and (4) must be initialized: the m values range between [1.5, 2.5]; e is a predefined threshold value equal to 10^{-5}; $r = 3$ is the total number of clusters for each class; $v(0) = [v_1 : v_r]$ is the initial cluster centers obtained from each training subset; the number of iterations t is set to 30; the shaping parameter γ in this paper is set by trying and testing different values, and it ranges between 7 and 10.

4- Optimized Training Data Set (OTD)

In Table V, it is noted that HED has the majority samples of 28.98% of the training data set, and PD has the minimum sample number of 9.09%, which represents the minority class of this set. However, this data is unbalanced, and the model's output will be more biased toward the majority class. To fix this problem, C-set & MFCM was implemented to produce a balanced data set. Therefore, after applying the clustering algorithm to all subsets, the original set has been reduced, and 60 balanced data samples out of 176 were collected from the fault group combinations, in which every fault has ten samples. These samples represent the (OTD) set that will be inputted to the optimizable MCSVM, as appears in Appendix A.

Moreover, it is seen in Fig. 2(a) that applying the C-set method and MFCM has helped achieve the principle of data reduction and provide clear representation of fault clusters by removing outliers from the training data set. Also, it is noted that the boundaries between clusters are separated, making it easier to create the separating hyperplanes when the optimizable MCSVM is used. Then again, the random distribution of the original data samples in Fig. 2(b) has made it extremely hard to find the faults' clusters. Accordingly, MCSVM cannot set the appropriate separating hyperplanes, and we cannot forget the enormous number of support vectors, which will create some issues in the classification process regarding speed, accuracy, and efficiency.

5- Clarification of the Analytical Process

To understand the analytical process in the data optimization section and how the (OTD) set is obtained, we provide the following explanation:

After pre-processing and dividing the data into training/testing and using the C-set method to separate the training set into 20 groups as shown in Tables IV and V, we take group A as an example; this group has three types of faults with a total number of samples equal to 98, as shown in Table AII, and these samples are distributed as follows: PD = 16, LED = 31, HED = 51. The severity of these faults can be identified by the measuring the concentration level of the dissolved gases in the transformer oil, as described in 2.1.1.

After that, MFCM is used to cluster this group by first initializing its parameters as follows:

- The cluster centers are initialized by taking three random samples (each sample represents one fault). The initial

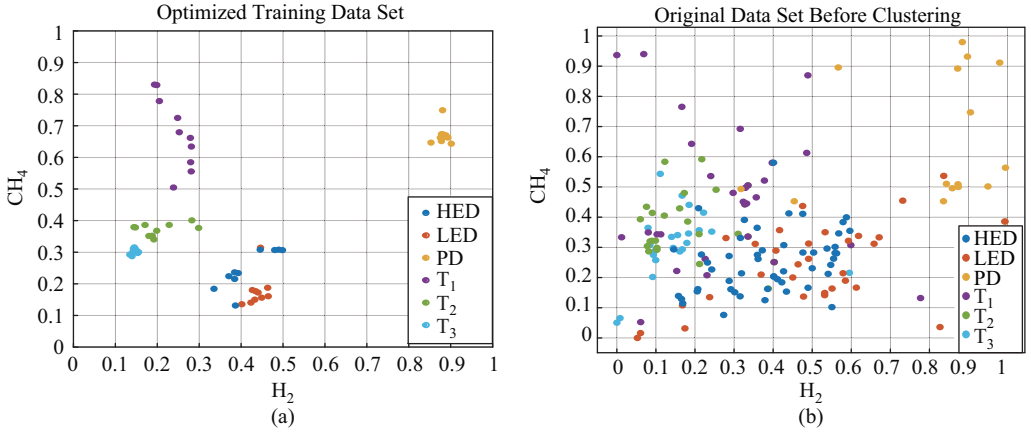


Fig. 2. (a) Optimized training data set after implementing the proposed model. (b) Original data before clustering.

cluster centers for group A are shown in Table VII.

- Then the parameters of MFCM are initialized as follows: $m = 2$ and $y = 10$, $r = 3$ (the number of clusters), $e = 10^{-5}$, $t = 30$ (iterations number).
- After the clustering is done, the algorithm output is being checked.
- In the MFCM output, if the samples are not correctly clustered or given a low membership value <0.5, they are considered as outliers, and they are removed.
- Then, the clustering technique is applied again to the remaining samples, and the values of m and y are modified accordingly to find the optimum results. In this case, the final values of m and y were 2.5 and 7, respectively.
- If samples in the second stage were correctly classified, then the final cluster centers are taken for this class.
- These steps are done to all 20 groups (individually), and the obtained cluster centers of each group are combined to form the (OTD) set.

TABLE VII
THE INITIAL CLUSTER CENTERS OF GROUP A

H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Diagnosis
0.94975	0.5016424	0.0004693	0.0004693	0.4974191	PD
0.17426	0.0316561	0.8586027	0.1062979	0.0034433	LED
0.56797	0.3461876	0.283085	0.3461876	0.0245399	HED

1) The Multiclass-SVM Classification Approach

In order to obtain an excellent fault classification in power transformers by using MCSVM, it is pivotal to build the best classification model by using the (OTD) set as well as tuning the model by setting specific parameters, such as box constraint level, kernel scale, and choosing the best optimizer, which is the Bayesian optimization algorithm in this paper. MATLAB 2019b is used for training and testing the optimizable MCSVM.

a) MCSVM Training: MCSVM hyper-parameter optimization and training are started using ten-fold cross-validation, the (OTD) in Table AI, and setting the model parameters in Table VIII.

TABLE VIII
MCSVM OPTIMIZABLE PARAMETERS

Type	Detail
Preset	Optimizable SVM
Kernel Function	Linear
Kernel Scale	Automatic
Multiclass Method	One-vs-One
Optimizer	Bayesian optimization
Acquisition Function	Expected improvement per second plus
Iterations	30

Moreover, the optimization performance of MCSVM, which has been conducted by using the Bayesian optimization algorithm, can be better explained by using the minimum classification error plot, as shown in Fig. 3.

In Fig. 3. the objective function of MCSVM is minimized by using the Bayesian optimization algorithm, which internally maintains a Gaussian process model of the objective function

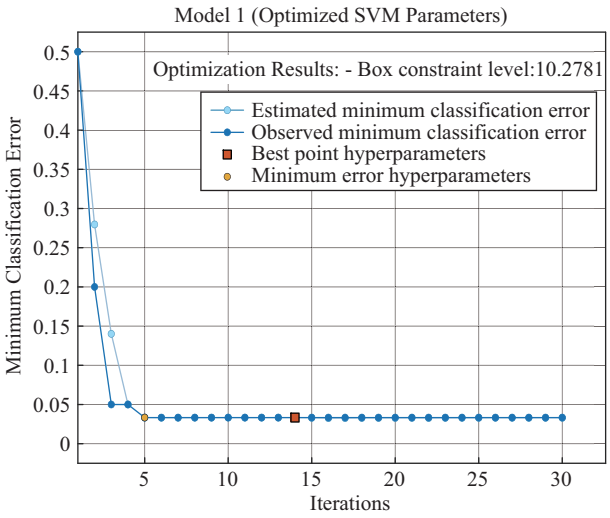


Fig. 3. The minimum classification error plot.

and chooses the set of hyper-parameter values that minimizes an upper confidence interval of the classification error objective model [25]. Then it uses the evaluation of the objective function to train the selected model. The minimum error or loss function obtained value in this process is 0.006. Moreover, the estimated minimum classification error in the light blue dots indicates the minimum error calculated during the optimization process when considering the hyper-parameters values at that iteration. The observed minimum classification error represented by the dark blue dots corresponds to the minimum classification error computed earlier by the optimization process. The best point hyper-parameter shown by the red square indicates the iteration that corresponds to the optimized hyper-parameter, which is equal to (box constraint level = 10.2781). Additionally, further explanation of the MCSVM prediction performance in terms of its confusion matrix (CM) is shown in Fig. 4.

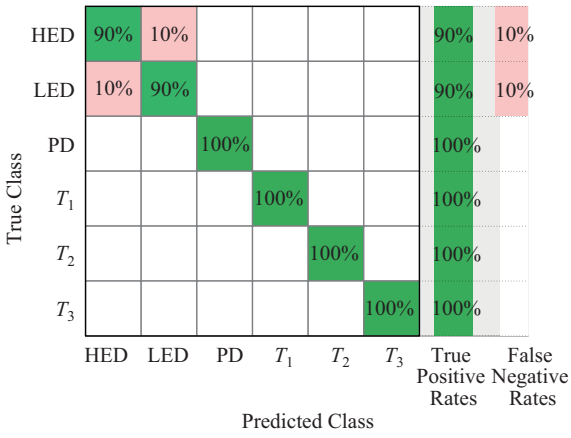


Fig. 4. Prediction performance of the optimizable MCSVM in the training process.

In this figure, the currently selected classifier has failed to classify only two samples; one is from a high energy discharge fault, and the other from Low energy discharge fault. Also, their true positive and false negative rates were 90% and 10%,

respectively. The other four defects were correctly classified with 100% true positive rate accuracy and a 0% false negative rate. Moreover, the training results of the optimized MCSVM are shown in Table IX.

TABLE IX
MCSVM TRAINING OUTPUT RESULTS

Accuracy	Total Misclassification cost	Prediction Speed	Training Time
96.70%	2	∼ 420 obs/sec	96.376 sec

D. Performance Evaluation of the Proposed Model

A total of 60 samples have been obtained from the MFCM algorithm and used to train the optimizable MCSVM. The trained model was imported to the MATLAB workspace for examination. In order to demonstrate our model's practicality, 30 testing samples were selected from the data set obtained from the Zibo electrical power company.

The results of examining the proposed model are shown in Fig. 5 in the form of (CM) four classes, which are LED, HED, T_1, T_3, which were correctly classified with 100% true positive and 0% false-negative percentages. The partial discharge (PD) and medium temperature overheating (T_2) classes have obtained (75.0%–80.0%) true positive and (25%–20%) false-negative rates respectively, in other words, the model has misclassified one partial discharge sample (PD) as a High temperature overheating fault (T_3). Also, one medium temperature sample (T_2) was misclassified as a low temperature overheating(T_1). Therefore, the overall accuracy obtained for this model is 93.3 %.

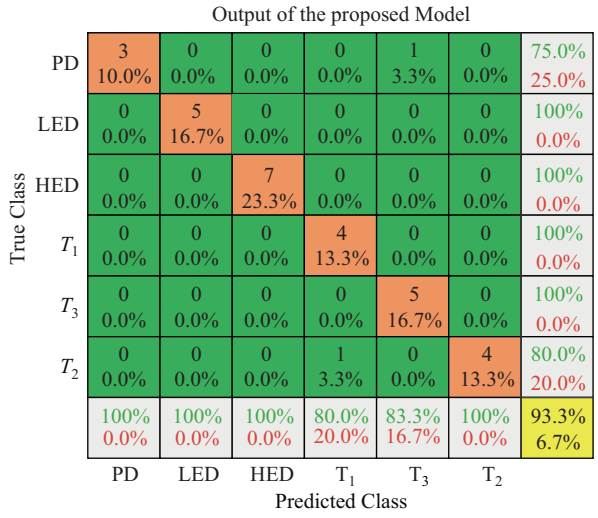


Fig. 5. Prediction output of C-set & MFCM-MCSVM model.

The final model performance can also be measured in terms of precision, sensitivity, and specificity for each type of fault, as shown in Table X.

III. DISCUSSION

To demonstrate our model's superior performance, which proves high consistency when used for classifying faults in the PT using the DGA data set. A detailed comparison in performance accuracy is performed between our model and other Artificial intelligent (AI) and conventional methods, such as (S1∼KNN) (K-nearest neighborhood), (S2∼SVM (cubic)), (S3∼Naive Bayes), (S4∼Decision Tree), (S5∼Fine Gaussian SVM), (S6∼Neural Networks (NN)), (S7∼MCSVM), (S8∼IEC 60599), (S9∼Rogers ratio), (S10 ∼Duval pentagon), and (S11 ∼Duval triangle). In this comparison, the original 176 data sets were used to train {S1, S2, S3, S4, S5, S6, S7}, and the 30 testing samples from the Zibo power company were used for testing these methods and other traditional methods. The classification performance accuracy output between our model (S0∼C-set & MFCM-MCSVM) and other models with respect to the original diagnosis (S) obtained from the Zibo power supply company in China is shown in Table AIV, where we can note that the (S1&S2&S7) methods have correctly classified 24 out of 30 samples with an accuracy of 80%. Moreover, the S3 method has classified only 20 samples out of 30 with an overall 67% accuracy. The S4 method has also managed to classify 20 samples correctly, with 67% total accuracy. In addition, the S5 method has classified 19 out of 30 samples correctly with 60% accuracy. The S6 method managed to classify only 18 out of 30 samples correctly with an overall accuracy of 60%. Moreover, as we move in the table from AI methods to traditional methods, it is noted that S8 managed to identify the power transformer's general status by identifying 26 out of 30 faulty samples as alert or alarm and the other four samples were misidentified as normal. However, our concern is not about identifying the general status of the transformer. Instead, we focus on determining the exact type of faults that occur inside the transformer, not to forget the causes of these faults may differ from one to another, which require different solutions from one fault to another. Therefore, the overall accuracy of S8 was considered as 0%. Moreover, the S9 method has also proved poor classification in which it managed to classify 11 out of 30 samples, which gives an accuracy of 37%. As for S10, it correctly classified 16 out of 30 samples with an accuracy of 53%. In addition, S11 has correctly classified 19 out of 30 with an accuracy of 63%; in samples, the numbers {24–27} of the Duval triangle doesn't give a succinct description of the defect. It shows that the defect is a combination of electrical and thermal faults in the same sample, while the correct finding is High temperature overheating (T_3) and Medium temperature overheating (T_2) for sample numbers 24 and 27, individually.

In addition, for a fair comparison between our model and other methods, all AI algorithms have been optimized using the Bayesian optimization method and trained with the

TABLE X
THE PERFORMANCE OF THE PROPOSED MODEL WITH RESPECT TO EACH FAULT TYPE

Proposed Model	Precision	Sensitivity	Specifity
PD	100%	75%	100%
LED	100%	100%	100%
HED	100%	100%	100%
T1	80%	100%	96%
T3	83%	100%	96%
T2	100%	80%	100%

TABLE XI
THE RESULTS OF USING THE OPTIMIZED TRAINING DATA WITH AI METHODS

Diagnosis Method	Training Accuracy	Testing Accuracy	Prediction Speed (obs/sec)	Training Time (sec)
S0	96.7%	0.933	∼ 420	96.37
S1	96.7%	0.67	∼ 750	105.7
S2	95.0%	0.63	∼ 320	117
S3	95.0%	0.43	∼ 890	155.5
S4	95.0%	0.5	∼ 1200	54.99
S5	97.0%	0.63	∼ 350	104.1
S6	98.0%	0.4	∼ 1400	29.19

optimized training data set. The training results of the AI algorithms are shown in Table XI.

In Table XI, it is noted that all AI models have obtained high training accuracy. However, these models tend to overfit and cause problems in the testing accuracy results. This table also compares the prediction speed and the training time of our model S0 and other models, in which the S6∼NN model has obtained the highest prediction speed and the fastest training time. Still, it overfits and fails to obtain high classification accuracy compared to our model. Since the C-set & MFCM is considered a data optimization method that involves performing multiple clustering processes to all subsets, the obtained optimized training data cannot be applied to the conventional techniques for comparison because most of the traditional methods work on the principle of taking the ratio of the input gases that contain the unprocessed data (data collected directly from the source) to obtain the final results. Another reason for not using the optimized data with conventional techniques is the range value of this data between $\{0, 1\}$, which is not suitable for these methods.

Figure 6 demonstrates a comparison in classification accuracy between our model, which obtained 93.3% accuracy, and other models.

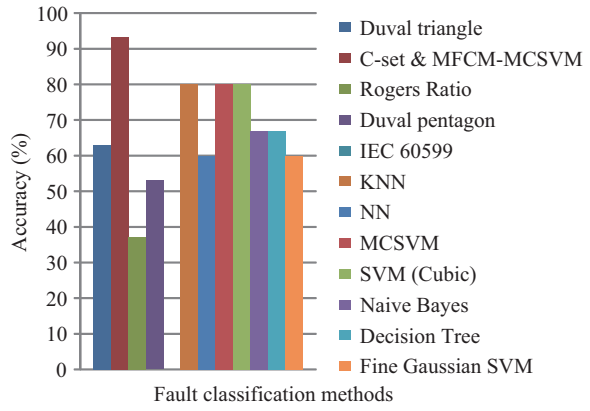


Fig. 6. A comparison between the accuracy of the proposed model and the accuracies of the other intelligent and conventional techniques.

In a future study, we will convert our model from off-line into online mode by adding a diagnostic monitor, such as DGA 900, into the transformer. This type of sensor uses photoacoustic spectroscopy technology to detect up to 9 gases inside the transformer oil. Moreover, the gas data will be transmitted wirelessly in real-time to the central station. After gas data are received, our model will optimize the data and obtain the final fault diagnosis.

IV. CONCLUSION

Due to the problems of imbalanced data sets, outliers, and boundary ratios associated with conventional and AI methods, this paper has introduced a new off-line model for classifying faults in the PT based on (C-set & MFCM-OMCSVM). This model consists of two parts. The first one is the data optimization step that involves using the C-set & MFCM to solve the previously mentioned problems and obtain the (OTD) set. The second one is using the (OTD) to train the optimizable MCSVM model.

The final classification results show that the (OTD) set has helped the MCSVM to successfully obtain higher generalization performance by characterizing the learner's ability to predict unlabelled data sets (testing data). Also, the optimizable MCSVM has successfully obtained the robust separating hyperplanes that separate the transformer fault classes, which can be seen by the low number of misclassified data samples in the testing mode. In addition, a comparison was performed between the accuracy of the proposed model and the accuracies of the other intelligent and conventional techniques, in which our model achieved the highest accuracy, which reached 93.3%.

This model's main disadvantage is the long time spent in the data preparation steps in terms of dividing the training data into different groups and afterward examining each group separately.

APPENDIX

TABLE AI
OPTIMIZED TRAINING DATA SET OF 3 SUBSET COMBINATIONS

Num	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6	Diagnosis
1	0.881	0.749	0.119	0.055	0.077	PD
2	0.893	0.663	0.177	0.095	0.066	PD
3	0.882	0.673	0.168	0.090	0.069	PD
4	0.878	0.674	0.167	0.090	0.069	PD
5	0.888	0.665	0.171	0.096	0.068	PD
6	0.853	0.647	0.179	0.101	0.067	PD
7	0.875	0.662	0.169	0.093	0.077	PD
8	0.901	0.643	0.189	0.104	0.064	PD
9	0.878	0.652	0.175	0.097	0.064	PD
10	0.888	0.670	0.174	0.089	0.067	PD
11	0.386	0.131	0.664	0.180	0.025	HED
12	0.371	0.224	0.431	0.308	0.036	HED
13	0.385	0.215	0.447	0.303	0.033	HED
14	0.385	0.236	0.421	0.306	0.036	HED
15	0.446	0.308	0.235	0.300	0.138	HED
16	0.490	0.308	0.243	0.301	0.143	HED
17	0.480	0.307	0.256	0.302	0.135	HED
18	0.499	0.307	0.271	0.305	0.111	HED
19	0.393	0.234	0.424	0.306	0.032	HED
20	0.335	0.184	0.475	0.306	0.027	HED
21	0.446	0.314	0.245	0.311	0.130	LED
22	0.449	0.156	0.635	0.170	0.038	LED
23	0.463	0.188	0.582	0.173	0.037	LED
24	0.427	0.180	0.561	0.183	0.076	LED
25	0.402	0.136	0.665	0.175	0.025	LED
26	0.423	0.141	0.663	0.170	0.026	LED
27	0.433	0.150	0.645	0.174	0.030	LED
28	0.465	0.161	0.631	0.165	0.039	LED

TABLE AI
OPTIMIZED TRAINING DATA SET OF 3 SUBSET COMBINATIONS (CONTINUED)

Num	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6	Diagnosis	Num	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6	Diagnosis
29	0.434	0.177	0.617	0.169	0.036	LED	45	0.144	0.379	0.154	0.388	0.041	T2
30	0.440	0.173	0.624	0.160	0.036	LED	46	0.180	0.351	0.144	0.365	0.038	T2
31	0.280	0.585	0.085	0.218	0.079	T1	47	0.192	0.340	0.132	0.397	0.040	T2
32	0.239	0.505	0.117	0.226	0.060	T1	48	0.148	0.378	0.171	0.363	0.031	T2
33	0.249	0.725	0.033	0.134	0.068	T1	49	0.228	0.386	0.163	0.308	0.035	T2
34	0.281	0.556	0.101	0.198	0.057	T1	50	0.299	0.377	0.137	0.242	0.018	T2
35	0.205	0.778	0.012	0.102	0.082	T1	51	0.145	0.315	0.058	0.555	0.072	T3
36	0.252	0.680	0.040	0.139	0.085	T1	52	0.150	0.306	0.057	0.560	0.077	T3
37	0.279	0.662	0.052	0.153	0.076	T1	53	0.155	0.301	0.063	0.559	0.073	T3
38	0.199	0.829	0.011	0.078	0.050	T1	54	0.140	0.288	0.057	0.586	0.066	T3
39	0.281	0.634	0.061	0.167	0.069	T1	55	0.143	0.311	0.053	0.559	0.077	T3
40	0.193	0.830	0.016	0.075	0.046	T1	56	0.153	0.298	0.060	0.572	0.071	T3
41	0.171	0.386	0.151	0.360	0.038	T2	57	0.148	0.310	0.087	0.521	0.053	T3
42	0.199	0.368	0.156	0.345	0.034	T2	58	0.146	0.301	0.060	0.568	0.070	T3
43	0.188	0.352	0.144	0.367	0.029	T2	59	0.142	0.301	0.070	0.563	0.063	T3
44	0.283	0.401	0.141	0.237	0.024	T2	60	0.135	0.293	0.055	0.581	0.068	T3

TABLE AII
THE DATA SET OF GROUP A

A (D1&D2&D3)													
Num	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6	Diagnosis	Num	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6	Diagnosis
1	0.884	0.980	0.000	0.000	0.020	PD	50	0.541	0.212	0.019	0.311	0.458	HED
2	0.897	0.932	0.007	0.006	0.056	PD	51	0.427	0.307	0.051	0.294	0.347	HED
3	0.872	0.892	0.002	0.001	0.105	PD	52	0.555	0.263	0.053	0.287	0.397	HED
4	0.980	0.912	0.000	0.003	0.085	PD	53	0.402	0.204	0.028	0.359	0.409	HED
5	0.950	0.502	0.000	0.000	0.497	PD	54	0.273	0.076	0.648	0.260	0.016	HED
6	0.567	0.896	0.000	0.001	0.103	PD	55	0.379	0.125	0.544	0.297	0.035	HED
7	0.993	0.386	0.169	0.386	0.060	PD	56	0.292	0.161	0.531	0.287	0.020	HED
8	0.843	0.510	0.465	0.025	0.000	PD	57	0.440	0.412	0.310	0.230	0.048	HED
9	0.995	0.564	0.336	0.100	0.000	PD	58	0.386	0.162	0.492	0.346	0.000	HED
10	0.859	0.496	0.318	0.187	0.000	PD	59	0.402	0.204	0.409	0.359	0.028	HED
11	0.318	0.493	0.320	0.187	0.000	PD	60	0.588	0.399	0.282	0.318	0.000	HED
12	0.905	0.747	0.137	0.107	0.009	PD	61	0.476	0.411	0.272	0.303	0.014	HED
13	0.873	0.509	0.407	0.084	0.000	PD	62	0.243	0.227	0.374	0.364	0.036	HED
14	0.836	0.453	0.181	0.364	0.002	PD	63	0.209	0.430	0.217	0.325	0.028	HED
15	0.874	0.500	0.414	0.086	0.000	PD	64	0.558	0.281	0.441	0.254	0.024	HED
16	0.454	0.453	0.181	0.364	0.003	PD	65	0.315	0.138	0.586	0.263	0.014	HED
17	0.551	0.163	0.603	0.209	0.025	LED	66	0.500	0.231	0.462	0.231	0.077	HED
18	0.732	0.455	0.182	0.182	0.182	LED	67	0.372	0.290	0.450	0.229	0.030	HED
19	0.613	0.167	0.650	0.183	0.000	LED	68	0.489	0.166	0.680	0.147	0.007	HED
20	0.478	0.137	0.733	0.120	0.011	LED	69	0.207	0.162	0.431	0.377	0.031	HED
21	0.168	0.108	0.751	0.130	0.011	LED	70	0.422	0.184	0.496	0.287	0.033	HED
22	0.532	0.142	0.472	0.089	0.297	LED	71	0.401	0.580	0.134	0.246	0.040	HED
23	0.837	0.537	0.171	0.146	0.146	LED	72	0.361	0.262	0.402	0.319	0.017	HED
24	0.579	0.214	0.568	0.189	0.029	LED	73	0.232	0.250	0.322	0.388	0.041	HED
25	0.671	0.333	0.570	0.095	0.002	LED	74	0.547	0.242	0.419	0.226	0.113	HED
26	0.658	0.312	0.396	0.212	0.081	LED	75	0.359	0.365	0.264	0.281	0.090	HED
27	0.585	0.190	0.578	0.211	0.021	LED	76	0.550	0.102	0.429	0.429	0.041	HED
28	0.619	0.338	0.541	0.108	0.014	LED	77	0.476	0.283	0.323	0.313	0.081	HED
29	0.053	0.000	0.701	0.299	0.000	LED	78	0.302	0.151	0.472	0.334	0.043	HED
30	0.369	0.210	0.583	0.196	0.011	LED	79	0.538	0.296	0.454	0.237	0.013	HED
31	0.476	0.437	0.294	0.183	0.086	LED	80	0.206	0.154	0.494	0.327	0.025	HED
32	0.491	0.313	0.334	0.181	0.173	LED	81	0.216	0.276	0.355	0.345	0.023	HED
33	0.827	0.036	0.655	0.237	0.072	LED	82	0.146	0.293	0.328	0.345	0.034	HED
34	0.061	0.016	0.839	0.113	0.032	LED	83	0.158	0.139	0.459	0.369	0.033	HED
35	0.174	0.032	0.859	0.106	0.003	LED	84	0.434	0.153	0.575	0.249	0.022	HED
36	0.407	0.289	0.361	0.325	0.024	LED	85	0.568	0.346	0.283	0.346	0.025	HED
37	0.416	0.357	0.357	0.286	0.001	LED	86	0.165	0.129	0.565	0.288	0.018	HED
38	0.279	0.331	0.064	0.113	0.491	LED	87	0.411	0.195	0.297	0.451	0.057	HED
39	0.237	0.135	0.277	0.138	0.450	LED	88	0.579	0.384	0.314	0.277	0.025	HED
40	0.993	0.386	0.169	0.386	0.060	LED	89	0.169	0.114	0.461	0.258	0.166	HED
41	0.464	0.244	0.085	0.091	0.580	LED	90	0.319	0.214	0.414	0.356	0.016	HED
42	0.354	0.311	0.040	0.100	0.548	LED	91	0.559	0.302	0.349	0.325	0.025	HED
43	0.450	0.200	0.049	0.232	0.519	LED	92	0.360	0.275	0.383	0.300	0.041	HED
44	0.533	0.350	0.068	0.142	0.439	LED	93	0.563	0.280	0.357	0.343	0.020	HED
45	0.532	0.149	0.032	0.212	0.607	LED	94	0.326	0.391	0.222	0.314	0.073	HED
46	0.491	0.262	0.068	0.160	0.510	LED	95	0.316	0.331	0.280	0.326	0.063	HED
47	0.593	0.322	0.068	0.085	0.526	LED	96	0.596	0.355	0.325	0.296	0.025	HED
48	0.504	0.283	0.065	0.237	0.415	HED	97	0.287	0.271	0.419	0.296	0.014	HED
49	0.427	0.221	0.062	0.348	0.369	HED	98	0.287	0.189	0.447	0.342	0.022	HED

TABLE AIII
THE TESTING DATA SET FROM ZIBO ELECTRICAL POWER COMPANY

Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Diagnos	Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Diagnos
1	0.928	0.939	0.000	0.000	0.061	PD	16	0.613	0.320	0.391	0.268	0.022	HED
2	0.950	0.868	0.004	0.004	0.124	PD	17	0.074	0.250	0.006	0.040	0.705	T1
3	0.901	1.000	0.000	0.000	0.000	PD	18	0.328	0.658	0.162	0.179	0.000	T1
4	0.237	0.013	0.232	0.755	0.000	PD	19	0.217	0.592	0.159	0.249	0.000	T1
5	0.520	0.278	0.389	0.181	0.153	LED	20	0.324	0.667	0.154	0.179	0.000	T1
6	0.268	0.120	0.648	0.193	0.040	LED	21	0.037	0.276	0.000	0.413	0.311	T3
7	0.063	0.012	0.932	0.050	0.006	LED	22	0.181	0.346	0.025	0.583	0.046	T3
8	0.181	0.049	0.767	0.168	0.017	LED	23	0.117	0.314	0.108	0.578	0.000	T3
9	0.525	0.146	0.621	0.209	0.024	LED	24	0.165	0.168	0.208	0.613	0.012	T3
10	0.569	0.236	0.031	0.318	0.415	HED	25	0.166	0.290	0.074	0.620	0.016	T3
11	0.462	0.226	0.047	0.346	0.381	HED	26	0.159	0.466	0.012	0.321	0.201	T2
12	0.324	0.192	0.074	0.307	0.426	HED	27	0.158	0.393	0.189	0.414	0.004	T2
13	0.482	0.373	0.029	0.374	0.225	HED	28	0.382	0.502	0.127	0.371	0.000	T2
14	0.432	0.201	0.478	0.321	0.000	HED	29	0.010	0.443	0.271	0.287	0.000	T2
15	0.503	0.242	0.444	0.284	0.030	HED	30	0.061	0.476	0.174	0.350	0.000	T2

TABLE AIV
PERFORMANCE COMPARISON BETWEEN THE MFCM-MSVM MODEL AND OTHER CLASSIFICATION TECHNIQUES

Num.	S	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
1	PD	PD	PD	PD	PD	PD	PD	PD	PD	Alert	NA	PD	PD
2	PD	PD	PD	PD	PD	PD	PD	PD	PD	Alert	PD	PD	PD
3	PD	PD	PD	PD	PD	PD	PD	PD	PD	Alert	NA	PD	PD
4	PD	T3	T3	T3	T2	T3	HED	LED	T3	Alert	NA	LED	LED
5	LED	LED	LED	LED	LED	LED	LED	LED	LED	Alarm	LED	LED	LED
6	LED	LED	HED	HED	LED	LED	HED	LED	LED	Alarm	NA	LED	LED
7	LED	LED	LED	LED	LED	T1	LED	LED	LED	Alarm	NA	LED	LED
8	LED	LED	LED	LED	LED	LED	LED	LED	LED	Alert	HED	LED	LED
9	LED	LED	LED	LED	LED	LED	LED	LED	LED	Alert	Normal	T1	LED
10	HED	HED	HED	HED	LED	HED	HED	HED	HED	Normal	NA	T1	T3
11	HED	HED	HED	HED	HED	HED	HED	HED	HED	Alert	NA	T1	T3
12	HED	HED	HED	HED	LED	LED	HED	HED	HED	Alert	T1	T1	T3
13	HED	HED	HED	HED	HED	HED	HED	HED	HED	Alarm	HED	HED	T2
14	HED	HED	HED	HED	T1	HED	HED	HED	HED	Alarm	HED	HED	HED
15	HED	HED	HED	HED	HED	HED	HED	HED	HED	Alert	HED	HED	HED
16	HED	HED	HED	HED	HED	HED	HED	HED	HED	Alert	HED	T1	HED
17	T1	T1	T1	T1	LED	LED	HED	HED	LED	Alert	PD	LED	T1
18	T1	T1	T1	T1	T1	T1	T1	T1	T1	Normal	T2	T1	T2
19	Tl	T1	T2	T2	T1	T1	HED	T1	T1	Alert	NA	T3	HED
20	T1	T1	T1	T1	T1	T1	HED	T1	T1	Alert	T2	T2	T2
21	T3	T3	T3	T3	T3	LED	HED	T2	‘T2’	Alarm	T3	T3	T3
22	T3	T3	T3	T3	T3	T3	T3	T2	‘T3’	Alarm	T3	T3	T3
23	T3	T3	T3	T3	T3	T3	T3	T1	‘T3’	Alarm	NA	T3	T3
24	T3	T3	T3	T1	T2	T3	HED	T2	‘T3’	Alarm	HED	HED	LED/T3
25	T3	T3	T3	T3	T3	T3	T3	T2	‘T3’	Alarm	NA	T3	T3
26	T2	T1	T1	T2	T1	T1	HED	T3	T1	Alert	NA	PD	T3
27	T2	T2	T1	T2	T2	T1	T1	T3	‘T3’	Alarm	T1	T3	LED/T3
28	T2	T2	T1	T1	T1	T1	T1	T3	T1	Normal	T2	T3	T2
29	T2	T2	T2	T2	T1	HED	HED	T3	T2	Normal	T2	Stray gassing oil <200c	T2
30	T2	T2	T2	T2	T1	T1	T2	T3	T2	Alert	T2	T2	T2

TABLE AV
THE ORIGINAL 239 DATA SET

Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS	Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS
1	0.461	0.215	0.615	0.158	0.012	Normal	17	0.520	0.361	0.417	0.222	0.000	Normal
2	0.335	0.383	0.228	0.174	0.215	Normal	18	0.367	0.456	0.070	0.474	0.000	Normal
3	0.838	0.645	0.258	0.097	0.000	Normal	19	0.468	0.693	0.160	0.141	0.005	Normal
4	0.461	0.215	0.615	0.158	0.012	Normal	20	0.465	0.850	0.060	0.090	0.000	Normal
5	0.337	0.045	0.500	0.418	0.037	Normal	21	0.288	0.422	0.115	0.446	0.017	Normal
6	0.468	0.693	0.160	0.141	0.005	Normal	22	0.264	0.732	0.117	0.149	0.002	Normal
7	0.234	0.119	0.184	0.650	0.047	Normal	23	0.429	0.168	0.116	0.716	0.000	Normal
8	0.372	0.210	0.152	0.566	0.072	Normal	24	0.000	1.000	0.000	0.000	0.000	Normal
9	0.000	0.471	0.294	0.235	0.000	Normal	25	0.000	0.154	0.454	0.393	0.000	Normal
10	0.255	0.274	0.654	0.038	0.034	Normal	26	0.230	0.216	0.340	0.444	0.000	Normal
11	0.000	0.263	0.357	0.246	0.134	Normal	27	0.000	0.331	0.190	0.401	0.078	Normal
12	0.471	0.519	0.370	0.111	0.000	Normal	28	0.000	0.247	0.141	0.548	0.064	Normal
13	0.275	0.297	0.081	0.595	0.027	Normal	29	0.433	0.235	0.627	0.139	0.000	Normal
14	0.523	0.509	0.340	0.151	0.000	Normal	30	0.504	0.449	0.322	0.230	0.000	Normal
15	0.599	0.493	0.368	0.139	0.000	Normal	31	0.000	0.310	0.502	0.188	0.000	Normal
16	0.897	0.397	0.214	0.254	0.135	Normal	32	0.867	0.438	0.336	0.227	0.000	Normal

TABLE AV
THE ORIGINAL 239 DATA SET (CONTINUED)

Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS	Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS
33	0.461	0.215	0.615	0.158	0.012	Normal	105	0.292	0.161	0.531	0.287	0.020	HED
34	0.928	0.939	0.000	0.000	0.061	PD	106	0.440	0.412	0.310	0.230	0.048	HED
35	0.950	0.868	0.004	0.004	0.124	PD	107	0.386	0.162	0.492	0.346	0.000	HED
36	0.901	1.000	0.000	0.000	0.000	PD	108	0.402	0.204	0.409	0.359	0.028	HED
37	0.840	0.116	0.094	0.790	0.000	PD	109	0.588	0.399	0.282	0.318	0.000	HED
38	0.884	0.980	0.000	0.000	0.020	PD	110	0.476	0.411	0.272	0.303	0.014	HED
39	0.897	0.932	0.007	0.006	0.056	PD	111	0.243	0.227	0.374	0.364	0.036	HED
40	0.872	0.892	0.002	0.001	0.105	PD	112	0.209	0.430	0.217	0.325	0.028	HED
41	0.980	0.912	0.000	0.003	0.085	PD	113	0.558	0.281	0.441	0.254	0.024	HED
42	0.950	0.502	0.000	0.000	0.497	PD	114	0.315	0.138	0.586	0.263	0.014	HED
43	0.567	0.896	0.000	0.001	0.103	PD	115	0.500	0.231	0.462	0.231	0.077	HED
44	0.993	0.386	0.169	0.386	0.060	PD	116	0.372	0.290	0.450	0.229	0.030	HED
45	0.843	0.510	0.465	0.025	0.000	PD	117	0.489	0.166	0.680	0.147	0.007	HED
46	0.995	0.564	0.336	0.100	0.000	PD	118	0.207	0.162	0.431	0.377	0.031	HED
47	0.859	0.496	0.318	0.187	0.000	PD	119	0.422	0.184	0.496	0.287	0.033	HED
48	0.318	0.493	0.320	0.187	0.000	PD	120	0.401	0.580	0.134	0.246	0.040	HED
49	0.905	0.747	0.137	0.107	0.009	PD	121	0.361	0.262	0.402	0.319	0.017	HED
50	0.873	0.509	0.407	0.084	0.000	PD	122	0.232	0.250	0.322	0.388	0.041	HED
51	0.836	0.453	0.181	0.364	0.002	PD	123	0.547	0.242	0.419	0.226	0.113	HED
52	0.874	0.500	0.414	0.086	0.000	PD	124	0.359	0.365	0.264	0.281	0.090	HED
53	0.454	0.453	0.181	0.364	0.003	PD	125	0.550	0.102	0.429	0.429	0.041	HED
54	0.520	0.278	0.389	0.181	0.153	LED	126	0.476	0.283	0.323	0.313	0.081	HED
55	0.268	0.120	0.648	0.193	0.040	LED	127	0.302	0.151	0.472	0.334	0.043	HED
56	0.063	0.012	0.932	0.050	0.006	LED	128	0.538	0.296	0.454	0.237	0.013	HED
57	0.181	0.049	0.767	0.168	0.017	LED	129	0.206	0.154	0.494	0.327	0.025	HED
58	0.525	0.146	0.621	0.209	0.024	LED	130	0.216	0.276	0.355	0.345	0.023	HED
59	0.551	0.163	0.603	0.209	0.025	LED	131	0.146	0.293	0.328	0.345	0.034	HED
60	0.732	0.455	0.182	0.182	0.182	LED	132	0.158	0.139	0.459	0.369	0.033	HED
61	0.613	0.167	0.650	0.183	0.000	LED	133	0.434	0.153	0.575	0.249	0.022	HED
62	0.478	0.137	0.733	0.120	0.011	LED	134	0.568	0.346	0.283	0.346	0.025	HED
63	0.168	0.108	0.751	0.130	0.011	LED	135	0.165	0.129	0.565	0.288	0.018	HED
64	0.532	0.142	0.472	0.089	0.297	LED	136	0.411	0.195	0.297	0.451	0.057	HED
65	0.837	0.537	0.171	0.146	0.146	LED	137	0.579	0.384	0.314	0.277	0.025	HED
66	0.579	0.214	0.568	0.189	0.029	LED	138	0.169	0.114	0.461	0.258	0.166	HED
67	0.671	0.333	0.570	0.095	0.002	LED	139	0.319	0.214	0.414	0.356	0.016	HED
68	0.658	0.312	0.396	0.212	0.081	LED	140	0.559	0.302	0.349	0.325	0.025	HED
69	0.585	0.190	0.578	0.211	0.021	LED	141	0.360	0.275	0.383	0.300	0.041	HED
70	0.619	0.338	0.541	0.108	0.014	LED	142	0.563	0.280	0.357	0.343	0.020	HED
71	0.053	0.000	0.701	0.299	0.000	LED	143	0.326	0.391	0.222	0.314	0.073	HED
72	0.369	0.210	0.583	0.196	0.011	LED	144	0.316	0.331	0.280	0.326	0.063	HED
73	0.476	0.437	0.294	0.183	0.086	LED	145	0.596	0.355	0.325	0.296	0.025	HED
74	0.491	0.313	0.334	0.181	0.173	LED	146	0.287	0.271	0.419	0.296	0.014	HED
75	0.827	0.036	0.655	0.237	0.072	LED	147	0.287	0.189	0.447	0.342	0.022	HED
76	0.061	0.016	0.839	0.113	0.032	LED	148	0.074	0.250	0.006	0.040	0.705	T1
77	0.174	0.032	0.859	0.106	0.003	LED	149	0.328	0.658	0.162	0.179	0.000	T1
78	0.407	0.289	0.361	0.325	0.024	LED	150	0.217	0.592	0.159	0.249	0.000	T1
79	0.416	0.357	0.357	0.286	0.001	LED	151	0.324	0.667	0.154	0.179	0.000	T1
80	0.279	0.331	0.064	0.113	0.491	LED	152	0.069	0.940	0.000	0.015	0.045	T1
81	0.237	0.135	0.277	0.138	0.450	LED	153	0.316	0.692	0.000	0.154	0.154	T1
82	0.993	0.386	0.169	0.386	0.060	LED	154	0.489	0.870	0.000	0.101	0.029	T1
83	0.464	0.244	0.085	0.091	0.580	LED	155	0.486	0.613	0.040	0.210	0.138	T1
84	0.354	0.311	0.040	0.100	0.548	LED	156	0.000	0.937	0.016	0.027	0.020	T1
85	0.450	0.200	0.049	0.232	0.519	LED	157	0.240	0.536	0.008	0.228	0.228	T1
86	0.533	0.350	0.068	0.142	0.439	LED	158	0.012	0.333	0.012	0.049	0.605	T1
87	0.532	0.149	0.032	0.212	0.607	LED	159	0.191	0.643	0.001	0.259	0.097	T1
88	0.491	0.262	0.068	0.160	0.510	LED	160	0.398	0.580	0.180	0.240	0.000	T1
89	0.593	0.322	0.068	0.085	0.526	LED	161	0.357	0.466	0.173	0.362	0.000	T1
90	0.569	0.236	0.031	0.318	0.415	HED	162	0.298	0.480	0.148	0.371	0.000	T1
91	0.462	0.226	0.047	0.346	0.381	HED	163	0.329	0.497	0.144	0.359	0.000	T1
92	0.324	0.192	0.074	0.307	0.426	HED	164	0.061	0.052	0.745	0.201	0.001	T1
93	0.482	0.373	0.029	0.374	0.225	HED	165	0.336	0.505	0.139	0.354	0.002	T1
94	0.432	0.201	0.478	0.321	0.000	HED	166	0.378	0.521	0.161	0.318	0.000	T1
95	0.503	0.242	0.444	0.284	0.030	HED	167	0.332	0.445	0.192	0.363	0.000	T1
96	0.613	0.320	0.391	0.268	0.022	HED	168	0.336	0.336	0.092	0.235	0.002	T1
97	0.504	0.283	0.065	0.237	0.415	HED	169	0.328	0.443	0.109	0.121	0.000	T1
98	0.427	0.221	0.062	0.348	0.369	HED	170	0.103	0.343	0.156	0.395	0.002	T1
99	0.541	0.212	0.019	0.311	0.458	HED	171	0.403	0.251	0.186	0.160	0.000	T1
100	0.427	0.307	0.051	0.294	0.347	HED	172	0.777	0.132	0.031	0.058	0.001	T1
101	0.555	0.263	0.053	0.287	0.397	HED	173	0.153	0.222	0.178	0.447	0.000	T1
102	0.402	0.204	0.028	0.359	0.409	HED	174	0.226	0.262	0.230	0.282	0.000	T1
103	0.273	0.076	0.648	0.260	0.016	HED	175	0.080	0.349	0.357	0.208	0.005	T1
104	0.379	0.125	0.544	0.297	0.035	HED	176	0.112	0.343	0.375	0.170	0.000	T1

TABLE AV
THE ORIGINAL 239 DATA SET (CONTINUED)

Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS	Num	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2	DiAGNOSIS
177	0.231	0.208	0.145	0.415	0.001	T1	209	0.403	0.251	0.160	0.186	0.000	T2
178	0.324	0.451	0.104	0.121	0.000	T1	210	0.211	0.344	0.126	0.253	0.067	T2
179	0.166	0.765	0.040	0.029	0.000	T1	211	0.122	0.405	0.189	0.284	0.001	T2
180	0.599	0.307	0.078	0.016	0.000	T1	212	0.037	0.276	0.000	0.413	0.311	T3
181	0.159	0.466	0.012	0.321	0.201	T2	213	0.181	0.346	0.025	0.583	0.046	T3
182	0.158	0.393	0.189	0.414	0.004	T2	214	0.117	0.314	0.108	0.578	0.000	T3
183	0.382	0.502	0.127	0.371	0.000	T2	215	0.165	0.168	0.208	0.613	0.012	T3
184	0.010	0.443	0.271	0.287	0.000	T2	216	0.166	0.290	0.074	0.620	0.016	T3
185	0.061	0.476	0.174	0.350	0.000	T2	217	0.222	0.415	0.007	0.521	0.057	T3
186	0.173	0.480	0.002	0.426	0.092	T2	218	0.085	0.308	0.018	0.578	0.095	T3
187	0.181	0.385	0.000	0.464	0.151	T2	219	0.080	0.365	0.000	0.469	0.166	T3
188	0.123	0.583	0.001	0.228	0.188	T2	220	0.097	0.288	0.093	0.619	0.000	T3
189	0.161	0.429	0.000	0.452	0.119	T2	221	0.185	0.441	0.011	0.541	0.008	T3
190	0.099	0.322	0.244	0.434	0.000	T2	222	0.595	0.216	0.108	0.588	0.088	T3
191	0.103	0.298	0.220	0.477	0.005	T2	223	0.167	0.471	0.012	0.411	0.105	T3
192	0.085	0.320	0.186	0.383	0.111	T2	224	0.000	0.051	0.001	0.441	0.507	T3
193	0.103	0.292	0.219	0.482	0.007	T2	225	0.092	0.202	0.011	0.676	0.111	T3
194	0.060	0.393	0.196	0.410	0.001	T2	226	0.111	0.543	0.003	0.354	0.100	T3
195	0.091	0.414	0.219	0.367	0.000	T2	227	0.144	0.298	0.020	0.593	0.089	T3
196	0.089	0.322	0.207	0.471	0.000	T2	228	0.243	0.352	0.101	0.496	0.051	T3
197	0.254	0.491	0.201	0.308	0.000	T2	229	0.156	0.341	0.075	0.497	0.088	T3
198	0.076	0.434	0.192	0.374	0.000	T2	230	0.008	0.066	0.049	0.820	0.066	T3
199	0.336	0.505	0.139	0.354	0.002	T2	231	0.099	0.258	0.013	0.626	0.103	T3
200	0.217	0.592	0.249	0.159	0.000	T2	232	0.211	0.357	0.005	0.554	0.085	T3
201	0.078	0.304	0.174	0.434	0.010	T2	233	0.179	0.315	0.100	0.578	0.007	T3
202	0.328	0.443	0.109	0.121	0.000	T2	234	0.167	0.293	0.106	0.600	0.000	T3
203	0.403	0.251	0.186	0.160	0.000	T2	235	0.184	0.346	0.078	0.575	0.001	T3
204	0.336	0.336	0.092	0.235	0.002	T2	236	0.161	0.286	0.121	0.583	0.010	T3
205	0.212	0.245	0.040	0.495	0.008	T2	237	0.140	0.335	0.071	0.591	0.004	T3
206	0.388	0.164	0.023	0.402	0.024	T2	238	0.093	0.275	0.095	0.627	0.003	T3
207	0.310	0.345	0.103	0.207	0.034	T2	239	0.145	0.295	0.114	0.588	0.003	T3
208	0.082	0.287	0.199	0.432	0.000	T2							

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